



GIRRAWEEEN HIGH SCHOOL

HALF YEARLY EXAMINATIONS

2017

MATHEMATICS

EXTENSION 1

Time Allowed: Two hours

(Plus 5 minutes reading time)

Instructions:

- There are 10 questions in this paper. All questions are compulsory.
- Use blue or black pen.
- Write all your answers in the Answer Booklets provided.
- For Questions 1 – 5, fill in the circle corresponding to the correct answer in your answer booklet.
- For Questions 6 – 10, start each question on a new page.
- Write on both sides of the paper.
- Show all necessary working.
- Board-approved calculators may be used.
- Mathematics reference sheets are provided.
- Marks may be deducted for careless or badly arranged work.
- Write 'End of Solutions' at the conclusion of your solutions to the task.

A. 3 B. 2 C. - 3 D. - 6

Question 6 (17 marks)

[3]

a. Solve the inequality $\frac{x+4}{x-3} \leq 2$.

b. The interval AB has endpoints $A(3, 2)$ and $B(4, 5)$. Find the coordinates of the point P which divides the interval AB externally in the ratio 3: 4. [3]

c. Find the value of m , where $m > 0$, such that the acute angle between the lines $y = 2x$ and $y = mx$ is 45° . [3]

d. (i) Express $\cos x - \sqrt{3}\sin x$ in the form $A\cos(x + \alpha)$ where $A > 0$ and α is an acute angle. [3]

(ii) Hence or otherwise, solve the equation [2]

$$\cos x - \sqrt{3}\sin x = 2 \text{ for } 0^\circ \leq x \leq 360^\circ.$$

e. Show that the expression $\frac{\sin 3\theta}{\sin \theta} - \frac{\cos 3\theta}{\cos \theta}$ is independent of θ . [3]

(where $\sin \theta \neq 0$ and $\cos \theta \neq 0$)

Question 7 (13 marks)

a. Consider the function $f(x) = \frac{e^x}{3 + e^x}$.

Note that e^x is always positive and that $f(x)$ is defined for all real x .

(i) Show that $f(x)$ has no stationary points. [2]

(ii) Show that $f''(x) = \frac{3e^x(3 - e^x)}{(3 + e^x)^3}$ [3]

(ii) Find the coordinates of the point of inflexion. [3]

b. Differentiate $\log_2 x^2$ [2]

c. Find the volume generated when $y = \log_e x$ is rotated about the y -axis between $y = 1$ and $y = 3$. Express your answer in exact form. [3]

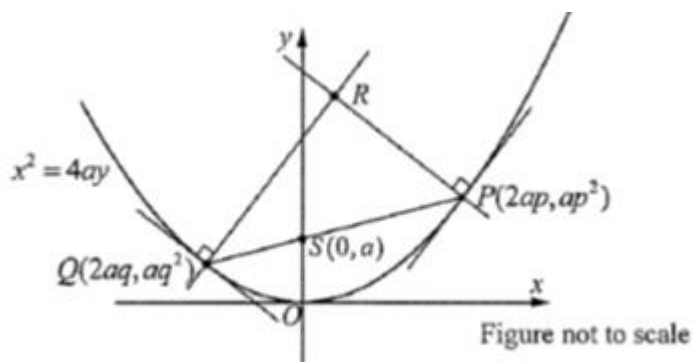
Question 8 (16 marks)

- a. When the polynomial $P(x) = (x - 1)(x - 2)Q(x) + 3x + k$ is divided by $(x - 1)$, the remainder is -1 . Find the remainder when $P(x)$ is divided by $(x - 2)$ [3]
- b. The equation $\tan^2 \theta + b \tan \theta + c = 0$ has roots $\tan \alpha$ and $\tan \beta$.
Find the expression for $\tan(\alpha + \beta)$ in terms of b and c . [2]
- c. Find the term independent of x in the expansion of $\left(3x^2 + \frac{2}{x}\right)^6$. [3]
- d. Five different fair dice are thrown together. Find the probability that
- (i) the five scores are all different [2]
 - (ii) the five scores include at most one 6. [3]
- e. In how many ways can 11 people occupy seats at two circular tables, where one table can seat 6 people and the other can seat 5 people. [3]

Question 9 (13 marks)

- a. The points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ are two points on the parabola $x^2 = 4ay$ such that PQ is a focal chord.

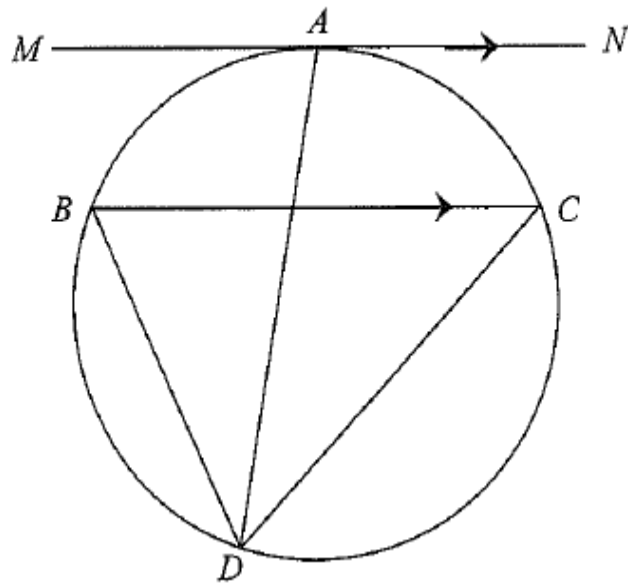
The normal at P and Q intersect at R .



- (i) Show that the equation of the normal at P is given by $x + py = ap^3 + 2ap$. [3]
- (ii) Show that R is the point $(-apq(p + q), a(p^2 + q^2 + pq + 2))$. [3]
- (iii) Hence, show that the equation of the locus of R as P and Q move on the parabola is given by $x^2 = a(y - 3a)$ [3]

Question 9 continues on the next page...

b.



In the diagram, MAN is tangent to the circle at A . BC is a chord of the circle such that $BC \parallel MN$.

D is a point on the circle.

(i) Copy the diagram into your answer booklet.

(ii) Show that AD bisects $\angle BDC$

[4]

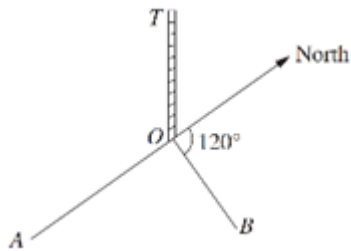
Examination continues on the next page...

Question 10 (9 marks)

a. Use the principal of Mathematical Induction to prove that

$$\frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \dots + \frac{n}{(n+1)!} = 1 - \frac{1}{(n+1)!} \quad \text{for } n \geq 1 \quad [4]$$

b.



From a point A due south of a tower, the angle of elevation of the top of the tower T , is 23° .
From another point B , on a bearing of 120° from the tower, the angle of elevation of T is 32° .
The distance AB is 200 metres.

(i) Copy the diagram into your writing booklet and add the given information. [1]

(ii) Find the height of the tower. [4]

End of Examination

MC

1. D 2. C 3. A 4. B 5. C

Q1

$$\tan \theta = \left| \frac{2 - (-1)}{1 + 2(-1)} \right| = 3 \quad \boxed{D}$$

Q2. A(-2, 5) B(4, -1)

$$\begin{array}{cc} 2 & 1 \\ \times & \times \end{array}$$

$$x = \frac{2(4) + 1(-2)}{3} ; y = \frac{2(-1) + 1(5)}{3} \quad \boxed{C}$$

Q3 $\cos(A-B) - \cos(A+B)$

$$= \cos A \cos B + \sin A \sin B - [\cos A \cos B - \sin A \sin B]$$

 $\boxed{A}$

Q4 8!

 $\boxed{B}$

Q5

Product of roots: $\alpha \cdot \frac{1}{\alpha} \cdot \beta = \beta = -\frac{6}{2}$ $\boxed{C}$

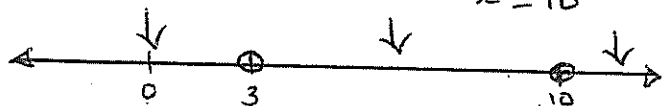
Question 6 (17 marks)

$$a. \frac{x+4}{x-3} \leq 2$$

$$x \neq 3 ; \text{ solve } \frac{x+4}{x-3} = 2$$

$$2x - 6 = x + 4$$

$$x = 10$$

Test $x=0$

$$\frac{4}{-3} \leq 2 \quad \checkmark$$

Test $x=5$

$$\frac{9}{2} \leq 2 \quad \times$$

Test $x=12$

$$\frac{16}{9} \leq 2 \quad \checkmark$$

 $\therefore$ Solution: $x < 3, x \geq 10$ $\boxed{3m}$

b. A(3, 2) B(4, 5)

$$\begin{array}{cc} -3 & 4 \\ \times & \times \end{array}$$

$$x = \frac{-3(4) + 4(3)}{-3 + 4} ; y = \frac{-3(5) + 4(2)}{-3 + 4}$$

P(0, -7)

 $\boxed{3m}$

$$c. y = 2x ; y = mx$$

$$m_1 = 2 ; m_2 = m$$

$$\tan 45^\circ = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$1 = \left| \frac{2 - m}{1 + 2m} \right|$$

$$1 + 2m = 2 - m \quad \text{or} \quad 1 + 2m = -(2 - m)$$

$$3m = 1$$

$$m = \frac{1}{3}$$

$$1 + 2m = -2 + m$$

$$m = -3$$

$$\therefore m = \frac{1}{3} \quad (\text{since } m > 0) \quad \boxed{3m}$$

$$d. (i) \cos x - \sqrt{3} \sin x = A \cos(x + \alpha)$$

$$= A \cos x \cos \alpha - A \sin x \sin \alpha$$

Equating coefficients,

$$A \cos \alpha = 1 \quad \text{--- ①} \quad A \sin \alpha = \sqrt{3} \quad \text{--- ②}$$

Squaring and adding ① & ②

$$A^2 (\cos^2 \alpha + \sin^2 \alpha) = 4$$

$$A^2 = 4, A = 2 \quad (A > 0)$$

$$\therefore 2 \cos \alpha = 1 ; 2 \sin \alpha = \sqrt{3} \quad \therefore \alpha < 90^\circ$$

$$\alpha = \frac{\pi}{3}$$

$$\therefore \cos x - \sqrt{3} \sin x = 2 \cos(x + \frac{\pi}{3}) \quad \boxed{3m}$$

$$ii) \cos x - \sqrt{3} \sin x = 2 \cos(x + \frac{\pi}{3}) = 2$$

$$\cos(x + \frac{\pi}{3}) = 1$$

$$x + \frac{\pi}{3} = 0, 2\pi, 4\pi$$

$$x = \frac{5\pi}{3}$$

 $\boxed{2m}$

$$e) \frac{\sin 3\theta}{\sin \theta} - \frac{\cos 3\theta}{\cos \theta}$$

$$= \frac{\sin 3\theta \cos \theta - \cos 3\theta \sin \theta}{\sin \theta \cos \theta}$$

$$= \frac{\sin(3\theta - \theta)}{\sin \theta \cos \theta} = \frac{\sin 2\theta}{\sin \theta \cos \theta} = \frac{2 \sin \theta \cos \theta}{\sin \theta \cos \theta}$$

= 2 which is independent of θ

a. $f(x) = \frac{e^x}{3+e^x}$

i) $f'(x) = \frac{(3+e^x)e^x - e^x \cdot e^x}{(3+e^x)^2}$

$$= \frac{3e^x + e^{2x} - e^{2x}}{(3+e^x)^2}$$

$$= \frac{3e^x}{(3+e^x)^2}$$

> 0 for all values of x
since $e^x > 0$

$\therefore f'(x)$ has no stationary points 2m

ii) $f''(x) = \frac{3e^x(3+e^x)^2 - 3e^x \cdot 2(3+e^x) \cdot e^x}{(3+e^x)^4}$

$$= \frac{3e^x(3+e^x)[3+e^x - 2e^x]}{(3+e^x)^4}$$

$$= \frac{3e^x(3-e^x)}{(3+e^x)^3} \quad [\text{GIVEN}]$$

Point of inflexion when $f''(x) = 0$

ie. $\frac{3e^x(3-e^x)}{(3+e^x)^3} = 0$

$$\therefore 3 - e^x = 0$$

$$e^x = 3$$

$$x = \ln 3$$

At $x = \ln 3$, $y = \frac{3}{3+3} = \frac{1}{2}$

Possible point of inflexion at $(\ln 3, \frac{1}{2})$

x	$\ln 2$	$\ln 3$	$\ln 4$
$f''(x)$	$\frac{4}{5^3} > 0$	0	$\frac{-7}{7^3} < 0$

There is a change in concavity at $(\ln 3, \frac{1}{2})$
is a point of inflexion 3m

$$f(x) = \frac{e^x}{3+e^x} < \frac{3+e^x}{3+e^x} = 1$$

$$\therefore f(x) < 1$$

$$\therefore 0 < f(x) < 1$$

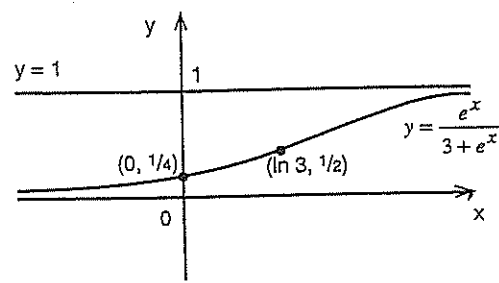
2m

iv) As $x \rightarrow \infty$, $f(x) \rightarrow 1$
As $x \rightarrow -\infty$, $f(x) \rightarrow 0$

$$\text{As } x \rightarrow -\infty, \quad e^x \rightarrow 0, \quad f(x) \rightarrow 0$$

2m

v)



b. $y = \log_2 x^2 = \frac{\log_e x^2}{\log_e 2}$
 $= \frac{1}{\log_e 2} \cdot 2 \log_e x$

$$\frac{dy}{dx} = \frac{1}{\log_e 2} \left[\frac{d}{dx} 2 \log_e x \right]$$

$$= \frac{1}{\log_e 2} \cdot \frac{2}{x}$$

$$= \frac{2}{x \log_e 2}$$

2m

c. $y = \log_e x \Rightarrow x = e^y \Rightarrow x^2 = e^{2y}$

$$V = \pi \int_1^3 x^2 dy$$

$$= \pi \int_1^3 e^{2y} dy$$

$$= \frac{\pi}{2} [e^{2y}]_1^3$$

$$= \frac{\pi}{6} [e^6 - e^2] \text{ cubic unit}$$

3m

Question 8 (10 marks)

a. $P(x) = (x-1)(x-2) \cdot Q(x) + 3x + k$

$$P(1) = 3(1) + k = -1$$

$$\therefore k = -4$$

$$\therefore P(x) = (x-1)(x-2) \cdot Q(x) + 3x - 4$$

$$P(2) = 3(2) - 4$$

$$= 2$$

$$\text{Remainder} = 2$$

3 marks 2m

b. $\tan^2 \theta + b \tan \theta + c = 0$

sum of roots: $\tan \alpha + \tan \beta = -b$

product of roots: $\tan \alpha \tan \beta = c$

$$\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$$

$$= \frac{-b}{1-c}$$

2m

c. $\left(3x^2 + \frac{2}{x}\right)^6$

$$T_{k+1} = {}^6C_k (3x^2)^{6-k} \left(\frac{2}{x}\right)^k$$

$$= {}^6C_k 3^{6-k} x^{12-2k} \cdot 2^k x^{-k}$$

$$= {}^6C_k 3^{6-k} 2^k x^{12-3k}$$

For term independent of x ,

$$12 - 3k = 0$$

$$k = 4$$

$$\therefore T_5 = {}^6C_4 3^2 2^4 = 2160$$

is the term independent of x .

3m

d(i) Total: 6^5

All 5 different: $6 \times 5 \times 4 \times 3 \times 2$

$$P(\text{all different}) = \frac{5}{54} \quad \boxed{2m}$$

ii) At most one 6 $\Rightarrow$ 1 six or 0 sixes

$$P(\text{at most 1 six}) = {}^5C_0 \left(\frac{1}{6}\right)^0 \left(\frac{5}{6}\right)^5 + {}^5C_1 \left(\frac{1}{6}\right)^1 \left(\frac{5}{6}\right)^4$$

$$= \left(\frac{5}{6}\right)^5 + 5 \left(\frac{5^4}{6^5}\right)$$

$$= 2 \left(\frac{5}{6}\right)^5$$

3m

e. ${}^6C_1 \times 5! \times 4!$ 3m

$\uparrow$ select 6 people
 $\uparrow$ arrange around table 1
 $\uparrow$ arrange around table 2.

Question 9 (13 marks)

a. i) $x^2 = 4ay$

$$y = \frac{x^2}{4a}$$

$$\frac{dy}{dx} = \frac{x}{2a}$$

At $P(2ap, ap^2)$, $m = \frac{2ap}{2a} = p$

$$\therefore m_{\text{normal}} = -\frac{1}{p}$$

$$E_{\text{normal}}: y - ap^2 = -\frac{1}{p}(x - 2ap)$$

$$py - ap^3 = -x + 2ap$$

$$x + py = ap^3 + 2ap \quad \text{--- (1)}$$

3m

ii) Similarly, E_{normal} at Q

$$x + qy = aq^3 + 2aq \quad \text{--- (2)}$$

(1) - (2)

$$(p-q)y = a(p^3 - q^3) + 2a(p-q)$$

$$= a(p-q)(p^2 + pq + q^2) + 2a(p-q)$$

$$y = a(p^2 + pq + q^2) + 2a$$

$$= a(p^2 + pq + q^2 + 2) \quad \text{--- (3)}$$

$$x + pa(p^2 + q^2 + pq + 2) = ap^3 + 2ap$$

$$x + ap^3 + apq^2 + ap^2q + 2ap = ap^3 + 2ap$$

$$x = -ap^2q - apq^2$$

$$= -apq(p+q)$$

$$\therefore R(apq(p+q), a(p^2 + pq + q^2 + 2))$$

3m

iii) PQ is a focal chord $\therefore pq = -1$

At R,

$$x = -apq(p+q); y = a(p^2 + pq + q^2 + 2)$$

$$x = a(p+q) \quad ; \quad y = a(p^2 + q^2 + 1)$$

$$\therefore p+q = \frac{x}{a} \quad ; \quad y = a\left(\left(\frac{x}{a}\right)^2 - 2pq + 1\right)$$

$$= a\left(\left(\frac{x}{a}\right)^2 + 2 + 1\right)$$

$$= a\left(\frac{x^2}{a^2} + 3\right)$$

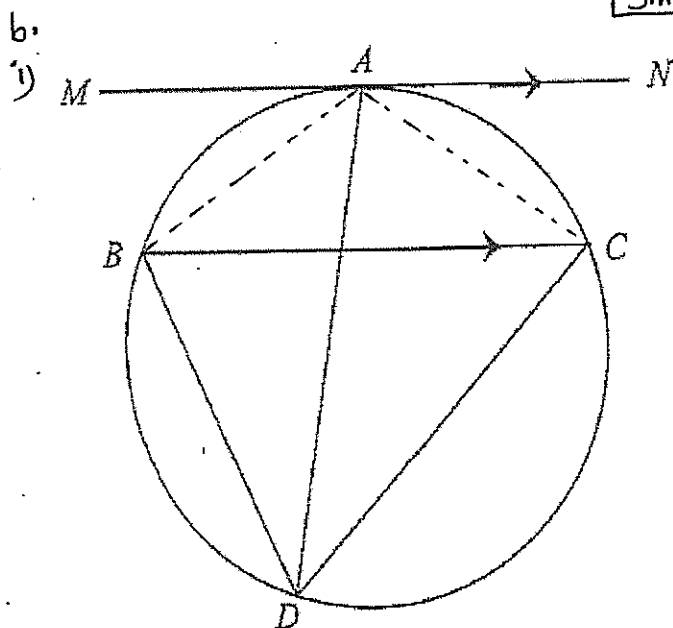
$$y = \frac{x^2 + 3a^2}{a}$$

$$ay = x^2 + 3a^2$$

$$x^2 = ay - 3a^2$$

$$\therefore x^2 = a(y - 3a)$$

3m



$\angle CDA = \angle CBA$ ($\angle$ s subtended by arc AC at the circumference)

$\angle CBA = \angle MAB$ (alternate $\angle$ s, $BC \parallel MN$)

$\angle MAB = \angle ADB$ ($\angle$ in the alternate segment)

$$\therefore \angle CDA = \angle ADB$$

$\therefore AD$ bisects $\angle BDC$.

4m

Question 10 (9 marks)

$$1. \frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \dots + \frac{n}{(n+1)!} = 1 - \frac{1}{(n+1)!}$$

for $n \geq 1$

Step 1: Show true for $n=1$

$$LHS = \frac{1}{2!} = \frac{1}{2} \quad ; \quad RHS = 1 - \frac{1}{(1+1)!} = \frac{1}{2}$$

$$LHS = RHS$$

$\therefore$ true for $n=1$

Step 2: Assume true for $n=k$

$$i.e. \frac{1}{2!} + \frac{2}{3!} + \dots + \frac{k}{(k+1)!} = 1 - \frac{1}{(k+1)!}$$

Step 3: Prove true for $n=k+1$

$$i.e. \frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \dots + \frac{k}{(k+1)!} + \frac{(k+1)}{(k+2)!} =$$

$$1 - \frac{1}{(k+2)!}$$

$$LHS = \frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \dots + \frac{k}{(k+1)!} + \frac{k+1}{(k+2)!}$$

$$= 1 - \frac{1}{(k+1)!} + \frac{k+1}{(k+2)!} \quad (\text{from step 2})$$

$$= 1 - \frac{k+2}{(k+2)(k+1)!} + \frac{k+1}{(k+2)!}$$

$$= 1 - \frac{k+2 - (k+1)}{(k+2)!}$$

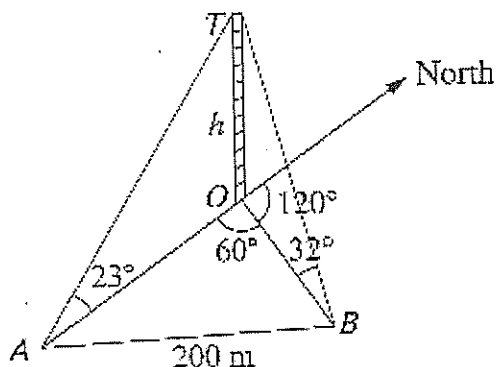
$$= 1 - \frac{1}{(k+2)!} = RHS.$$

also true for $n = k + 1$.

Step 4: By the principle of Mathematical Induction, the result is true for $n \geq 1$.

4m

b. i)



1m

ii) let height of tower $OT = h$

$$\text{In } \triangle OAT, \tan 23^\circ = \frac{h}{OA}$$

$$\therefore OA = \frac{h}{\tan 23^\circ}$$

$$\text{In } \triangle OBT, \tan 32^\circ = \frac{h}{OB}$$

$$\therefore OB = \frac{h}{\tan 32^\circ}$$

In $\triangle AOB$, using cosine rule

$$200^2 = OA^2 + OB^2 - 2 \times OA \times OB \cos 60^\circ$$

$$= \frac{h^2}{\tan^2 23^\circ} + \frac{h^2}{\tan^2 32^\circ} - \frac{2h^2}{\tan 23^\circ \tan 32^\circ} \cdot \frac{1}{2}$$

$$= h^2 \left(\frac{1}{\tan^2 23^\circ} + \frac{1}{\tan^2 32^\circ} - \frac{1}{\tan 23^\circ \tan 32^\circ} \right)$$

$$= h^2 (4.340 \dots)$$

$$h^2 = \frac{200^2}{4.340 \dots}$$

$$h = 95.99 \dots$$

$$h = 96 \text{ m (to the nearest m)}$$

4m